

Die Changeover & Storage Optimization at Johnson Controls International

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Industry Partners: Euvaldo Villaboim and Brady Amox, Johnson Controls International

Abstract

The die changeover and storage optimization project aimed to address areas for improvement in die storage and retrieval practices within the sheet metal production line at Johnson Controls International (JCI). A key challenge was prolonged changeover times between machine presses, leading to daily idle times, disruptions to the assembly line, and potential delays in client orders. Our team analyzed the current storage layout and retrieval processes, developed multiple solution options, and recommended a strategic layout and labeling system to minimize die travel distance and improve identification. The proposed solution is expected to reduce emissions (from reduced travel time), implement ergonomic considerations into labeling practices, and enhance customer satisfaction. Overall, the economic analysis indicates a positive return on investment, though continued data collection and lasting impacts of the implementations are considerations.

1 Introduction to the Company

Johnson Controls International (JCI) is a multinational corporation dedicated to efficient, sustainable solutions through advanced manufacturing and various mediums. With locations in over 150 countries worldwide, JCI has established a Norman, OK flagship location, known as the “Rooftop Center of Excellence,” with a focus on HVAC system production and innovation.

The Norman facility capstone team, composed of three students from The University of Oklahoma’s School of Industrial & Systems Engineering, partnered with JCI to address areas

for improvement in the die storage and changeover processes. These challenges lead to some prolonged downtime in between active hydraulic pressing, potentially affecting production times. The 900,000 square foot facility relies heavily on time-sensitive operations, such as die retrieval and the exchange of dies onto the press.

Our faculty advisor, Dr. Cesar Ruiz, and company contacts, Brady Amox (Industrial Engineer) and Euvaldo Villaboim (Director of Manufacturing Engineering), assisted us in identifying root causes and proposing data-driven solutions. Our team conducted measurement studies, gathered floor-level operational observations, and analyzed historical die usage data to optimize the storage layout for increased efficiency.

2 Problem Definition

The key focus addressed is the downtime during die changeover in JCI’s sheet metal cutting process, a crucial step for overall HVAC unit production. Two main contributing factors were identified: (1) non-optimized die storage, which extended retrieval times, and (2) occasional misplacement of dies within the storage system, which made them more difficult to locate during changeover.

Specifically, while the facility maintained an organizational system, it was not fully aligned with usage patterns or machine proximity. As a result, operators spent additional time locating certain dies, leading to avoidable delays. These observations led to the development of a labeling and zoning strategy to improve visibility and reduce retrieval errors.

We aim to: (1) reduce non-value-added time spent transporting dies by minimizing travel distance between die storage locations and machines and (2) implement practices to better identify and categorize dies based on usage frequency. Success is measured by reductions in non-value-added time spent transporting dies.

3 Problem Analysis

One of the primary workflow challenges identified in JCI’s sheet metal production process is the combination of some inefficient changeover practices and non-optimized storage of dies (Figure 1). Our analysis was conducted through a series of direct facility observations, measurements of travel distances between all possible storage locations and machine presses, die usage frequency analysis based on historical data, and input from floor operators and stakeholders. Due to the semi-irregular nature of die changes during shifts, we ultimately did not rely on time studies to inform our decision-making. Instead, we gathered more consistent data through distance measurements and usage trends to guide layout redesign.

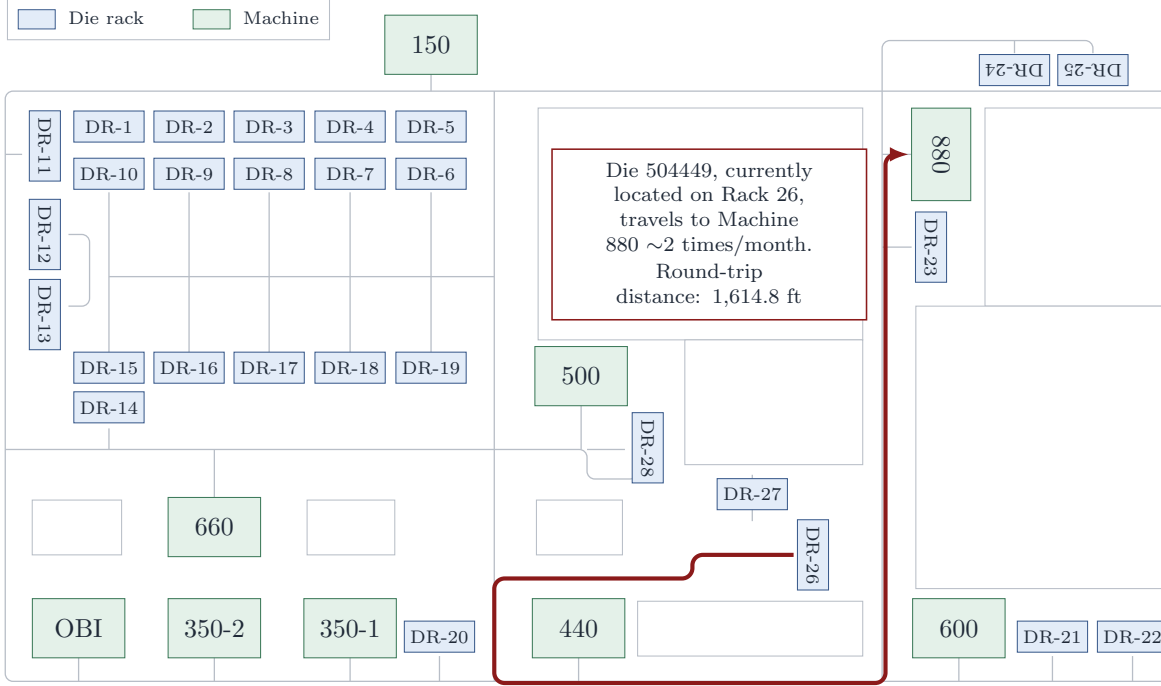


Figure 1: Current die storage layout example. Die 504449 is stored on Rack 26, travels to Machine 880 roughly twice per month, and the round-trip distance is 1,614.8 ft.

As part of our research, we explored academic work on storage policy optimization by [Lee and Elsayed \(2005\)](#), which emphasizes the value of assigning items based on demand and minimizing travel costs in constrained environments. We also examined a recent study on ergonomic warehouse practices ([Nasir et al., 2025](#)), which encouraged systems that support not only operational efficiency but also employee well-being and cognitive ease. These concepts shaped our interpretation of existing workflows and informed our final recommendations.

One notable insight was the absence of a placement strategy that directly linked die frequency of use to their proximity to machines. Instead, dies were stored wherever space allowed, leading to some inefficiencies in retrieval and occasional misplacement. There was not really a visual auditing system in place to ensure consistent storage adherence or to immediately flag unusual placement patterns. Using visualization tools such as spaghetti diagrams and facility maps created in Canva, along with distance matrix calculations, we highlighted several opportunities to consolidate and streamline die storage based on usage frequency and destination machine.

4 Decision Analysis

Based on stakeholder input and our technical evaluation, we considered several solution approaches. One focused on reorganizing die storage so that frequently used dies would be positioned closest to their respective machines. Another solution involved redesigning the labeling system with ergonomic and visual enhancements to improve clarity and usability. The final solution, which we ultimately selected, was a hybrid approach combining strategic die placement with the new labeling system. This hybrid strategy best addressed the challenges identified while remaining feasible for implementation. It met the key criteria of reducing average changeover time, supporting employee adoption, limiting cost, and requiring minimal disruption during rollout.

Selection criteria included effectiveness in reducing non-value-added time spent transporting dies, feasibility, cost, training needs, and maintainability. The hybrid solution was chosen for maximizing impact with minimal disruption.

5 Solution Process

Our team considered a few different approaches to solving this problem. The first was to measure current state and success by taking time studies. We initially planned to use facility layout improvement methods such as Simio and ARENA to simulate different layouts. Additionally, we considered making administrative changes such as changing scheduling methods, reassigning the set-up team, and improving order tracking. However, because of the number of dies, shelves, and machines, we opted to pursue a solution using linear programming.

Our final approach included both a new storage layout algorithm and a labeling system. The optimal storage layout was determined using linear programming to make an optimization model which minimized total distance travelled while transporting dies during a certain time period. This objective function was made up of a binary variable (1 if die d is assigned to shelf s , 0 otherwise), a distance parameter (the distance from shelf s to machine m), and a frequency parameter (frequency that die d is used by machine m during a period):

$$\min \sum_{d \in D} \sum_{s \in S} \sum_{m \in M} x_{ds} \cdot f_{md} \cdot d_{ms} \quad (1)$$

The full mathematical formulation is given in Appendix A. We designed the problem in this way so that the time period for frequency could be adjusted to the partner company's preference.

This model also takes into account the various constraints within the problem. These

constraints are that the number of dies that can fit on a shelf is related to the size of the die, the extra-large dies must be accessible by the die cart (which means they can only be stored on the second or third shelf), and that each die belongs to only one shelf. Once we had established the problem, we solved this linear optimization model using Python (Appendix B). The code pulls from Excel sheets which contain the distances between shelves and machines and the frequency of die usage by each machine for a trial three-month period. The solution prints out a phrase “Die ___ is located on shelf ___”, where the first blank has the die number (e.g., 504221) and the second blank is the rack and shelf number (e.g., R1_S3).

Additionally, we proposed an improved labeling system for the dies and shelves. The racks would be color coded in zones based on location on the floor. For example, all the shelves located in the “horseshoe,” which is the term that JCI uses to refer to the U-shaped die storage area, would be labelled as blue. Then, all of the dies that belong in that area would have blue labels on them. A map of the zones is shown in Figure 2. This would improve immediate recognition and quick visual audits. The labels placed on the dies would use a visual recognition tool called chunking to make the die numbers easier to read (e.g., “504 221” instead of “504221”). The label would also include the transportation method used for the die (forklift or die cart), routed machine, and the location of the die. An example of the ergonomically designed label is shown in Figure 3.

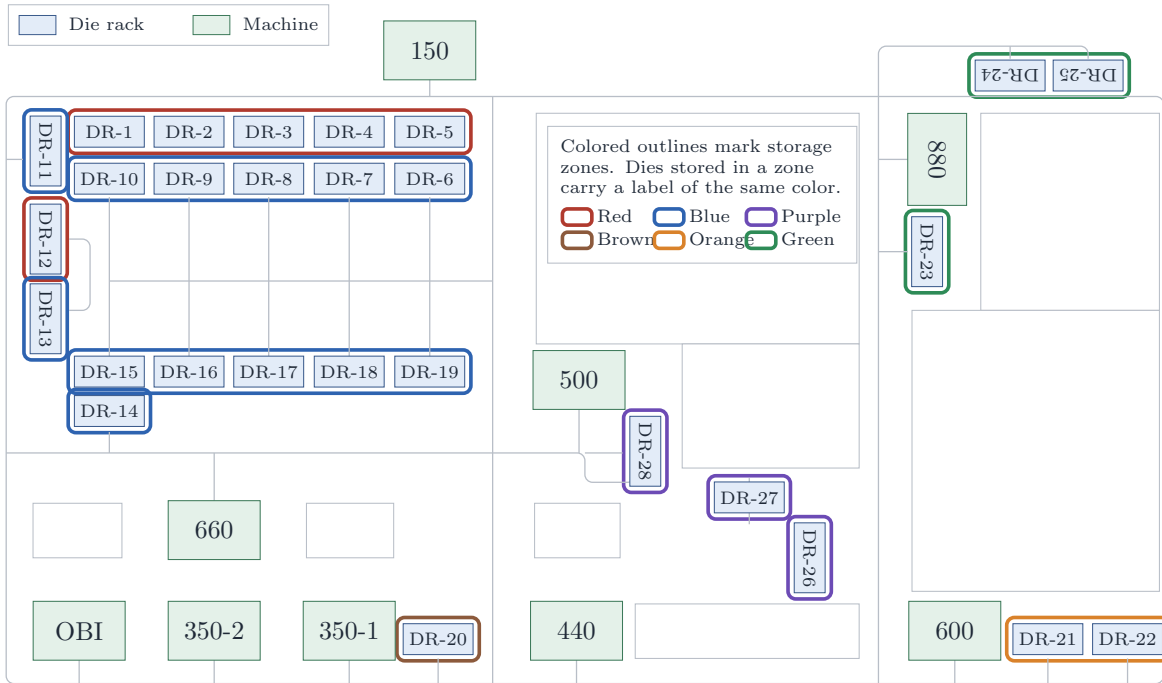


Figure 2: Die rack zoning by color.

Though this proposed solution is not the one that our team initially pursued, it meets the

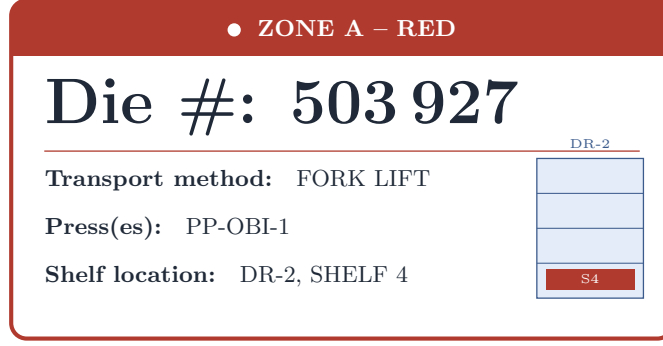


Figure 3: Proposed die label example, with chunked die number, transport method, routed press, and shelf location.

success measures defined at the start of the project. The possible improvement is significant and would benefit the company.

6 Economic Analysis and Systems Impacts

Table 1: Economic Analysis of Solution Options

	Distance (mi), 3 months	Non-value-added time (hrs), 3 months	Cost est. (\$), 3 months	Cost est. (\$), 1 year
Current	47.71	11.93	274.34	1,097.35
Future	32.38	8.10	186.19	744.74
Difference	15.33	3.83	88.15	352.61

We estimated ROI using the current layout vs. the proposed layout (Table 1), factoring in the round-trip transportation distance of all of the dies used in the given 3-month period with each layout, travel time assuming a forklift or die cart travels at approximately 4 mph (or 15 minutes per mile traveled of non-value-added time), and hourly labor costs for this travel time (estimated at \$23/hr). The economic analysis showed that the proposed hybrid solution leads to a significant reduction in non-value-added time, leading to saved time costs in a three-month-long period, and greater savings in a year-long period.

Assuming that it would take a single employee one nine-hour workday to rearrange all the dies in the new layout (at a rate of \$23/hr), the annual ROI for implementing this layout is approximately 70% (Table 2). For the new labeling system implementation, assuming that it can prevent a single employee (working at the same rate) at least one 15-minute missed

Table 2: Estimate of Annual Return on Investment (ROI)

Improvement type	Layout	Ergonomic	Total
Net return on investment (\$)	145.61	41.00	186.61
Cost of investment (\$)	207.00	28.00	235.00
ROI for 1 year (%)	70.34	146.43	79.41

trip a month (a missed trip meaning failure to retrieve and transport the correct die for any reason), and that it takes one employee approximately an hour to print off new labels (using approximately \$5 worth of materials), the annual ROI is about 146%. Together, the hybrid solution results in an ROI of approximately 80% per year.

7 Project Schedule

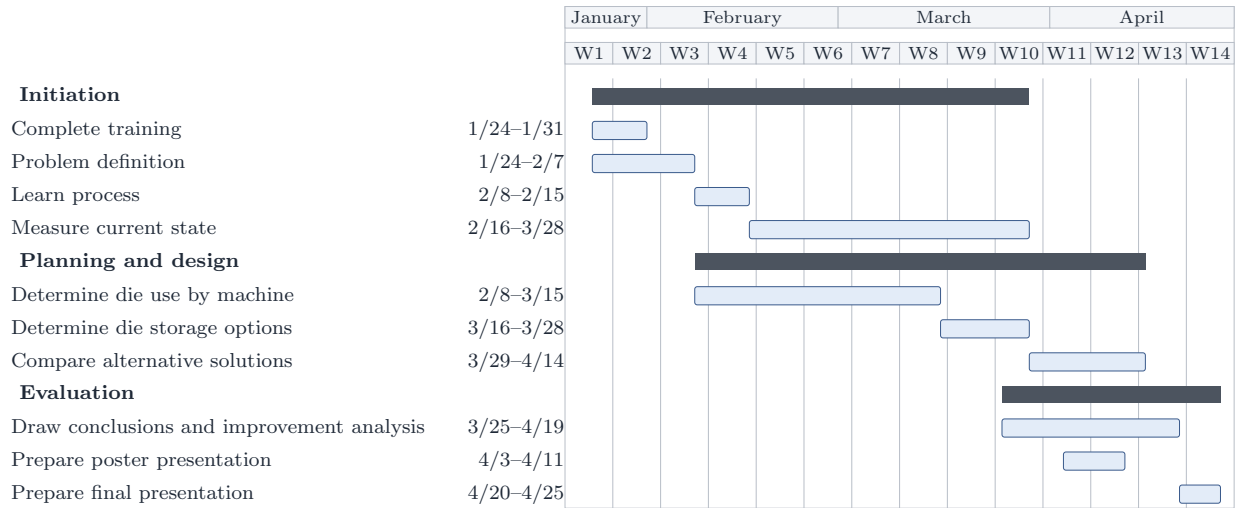


Figure 4: Project schedule (Gantt chart).

The original project schedule was developed in Gantt format using Excel (Figure 4), showing key milestones and deadlines. The actual schedule included deviations due to data collection scheduling issues as die changes happened semi-unpredictably, which led to adjustments in our measurement strategy. Shifting our project scope from die changeover time tracking to travel distance metrics required additional data collection and analysis. The project was actively managed using Teams, OneDrive, and regular updates through emails and text messages.

8 Summary, Conclusions, and Recommendations

The project successfully addressed areas for improvement in die storage and retrieval. The hybrid solution reduced travel distances and thus non-value-added transportation time, which is also supported by ergonomic improvements in die labeling. The economic analysis showed that the implementation of both the layout change and the labeling implementation would lead to a significant ROI. We recommend periodic audits of die placement, continued usage tracking, and training for all operators. Future implementations should include expanded data windows for die usage frequency analysis utilizing our code, as it could potentially be more beneficial to rearrange the dies periodically. We would also urge the re-utilization of the Python code based on scheduled die usage for upcoming parts in order to keep the layout of the dies more relevant to future orders, rather than only rearranging them based on historical orders that have already been completed.

9 Limitations and Reservations

Some limitations of this project included the time utilized for data collection, as this was a 5-month project accomplished by three students. A longer data collection period may be necessary to fully prove the lasting impacts of layout readjustment. This project also does not address any hydraulic press delays due to necessary coil changes, which also heavily impact production timing, specifically within the metal cutting process. Potential ripple effects of these implementations on other departments should be monitored.

References

- Lee, M.-K., and Elsayed, E. A. (2005). Optimization of warehouse storage capacity under a dedicated storage policy. *International Journal of Production Research*, 43(9):1785–1805. <https://doi.org/10.1080/13528160412331326496>
- Nasir, D., Venkitasubramony, R., and Jakhar, S. K. (2025). Ergonomics in warehouse design and operations: A systematic literature review. *Operational Research*, 25, Article 10. <https://doi.org/10.1007/s12351-024-00892-z>

A Mathematical Model of the Problem

Sets.

- $M = \{1, \dots, 9\}$: machines (e.g., OBI, 150, ..., 880)
- $S = \{1, \dots, 140\}$: shelves
- $D = \{1, \dots, 135\}$: dies

Subsets.

- $S_{xl} \subseteq S$: shelves for extra-large dies
- $D_{xs}, D_{sm}, D_{med}, D_{xl} \subseteq D$: die types by size

Parameters.

- d_{ms} : distance from machine m to shelf s
- f_{md} : frequency die d is used by machine m
- $xs_d = 5, sm_d = 3, med_d = 2$: shelf capacity by die size

Decision variable.

$$x_{ds} = \begin{cases} 1 & \text{if die } d \text{ is assigned to shelf } s \\ 0 & \text{otherwise} \end{cases} \quad \forall d \in D, \forall s \in S$$

Objective. Minimize the total transport cost based on how frequently each die is used by each machine and the distance between them:

$$\min \sum_{d \in D} \sum_{s \in S} \sum_{m \in M} x_{ds} \cdot f_{md} \cdot d_{ms}$$

Constraints.

$$\begin{aligned}
\sum_{d \in D_{xs}} x_{ds} &\leq xs_d & \forall s \in S \\
\sum_{d \in D_{sm}} x_{ds} &\leq sm_d & \forall s \in S \\
\sum_{d \in D_{med}} x_{ds} &\leq med_d & \forall s \in S \\
\sum_{d \in D} x_{ds} &\leq 1 & \forall s \in S \\
\sum_{s \in S} x_{ds} &= 1 & \forall d \in D \\
x_{ds} &= 0 & \forall d \in D_{xl}, \forall s \notin S_{xl}
\end{aligned}$$

B Python Code for the Optimization Model

The model is solved with Google OR-Tools' CP-SAT solver, reading the shelf-machine distance matrix and the die-machine usage frequency matrix from the Excel workbook (JCI_Data_python.xlsx).

```

from ortools.sat.python import cp_model
import openpyxl as opxl
import numpy as np
# Create the model
model = cp_model.CpModel()
# Load Excel workbook
file = opxl.load_workbook("/Users/samanthalynn/Desktop/JCI_Data_python.xlsx")
### SETS ###
M = list(file["distance"].cell(row = 1, column = m + 1).value for m in range(1,10))
    # Machines
S = list(file["distance"].cell(row = r + 1, column = 1).value for r in range(1, 141))
    # Shelves
D = list(file["frequency"].cell(row = d + 1, column = 1).value for d in range(2, 137))
    # Dies
machine_index = {m: i for i, m in enumerate(M)}
shelf_index = {s: i for i, s in enumerate(S)}
die_index = {d: i for i, d in enumerate(D)}
# Shelf subset that fits extra-large dies (2nd and 3rd shelves)
S_xl = set(file["die_category"].cell(row = 6, column = a + 1).value for a in range(1,
29))
# Die subsets categorized by size
D_xs = set(file["die_category"].cell(row = 1, column = a + 1).value for a in range(1,
24))
D_sm = set(file["die_category"].cell(row = 2, column = a + 1).value for a in range(1,
19))
D_med = set(file["die_category"].cell(row = 3, column = a + 1).value for a in range(1,

```

```

81))
D_xl = set(file["die_category"].cell(row = 4, column = a + 1).value for a in range(1,
20))
### PARAMETERS ###
dist_matrix = []      # Matrix of distances from each shelf to each machine
for row in sheet_dist.iter_rows(min_row = 2, max_row = 142, min_col = 2, max_col = 10):
    dist_matrix.append([cell.value for cell in row])
dist_matrix = np.array(dist_matrix).T # Shape: [machine][rack]
freq_matrix = []      # Matrix of historical frequency that each die was used by each
                        machine
for row in sheet_freq.iter_rows(min_row = 2, max_row = 137, min_col = 2, max_col = 10):
    freq_matrix.append([cell.value for cell in row])
freq_matrix = np.array(freq_matrix).T # Shape: [machine][die]
# Max number of dies allowed on a shelf by size
xs_limit = 5
sm_limit = 3
med_limit = 2
### DECISION VARIABLES ###
x = {} # x[d, s] = 1 if die d assigned to shelf s, 0 otherwise (this is a binary
        variable)
for d in D:
    for s in S:
        x[d, s] = model.NewBoolVar(f'x_{d}_{s}')
### CONSTRAINTS ###
# Each die assigned to exactly one shelf
for d in D:
    model.Add(sum(x[d, s] for s in S) == 1)
# Shelf capacity limits for each die type
for s in S:
    model.Add(sum(x[d, s] for d in D_xs if d in D) <= xs_limit)
    model.Add(sum(x[d, s] for d in D_sm if d in D) <= sm_limit)
    model.Add(sum(x[d, s] for d in D_med if d in D) <= med_limit)
    model.Add(sum(x[d, s] for d in D if d not in D_xs | D_sm | D_med) <= 1)
# Extra-large dies only go on S_xl shelves
for d in D_xl:
    for s in S:
        if s not in S_xl:
            model.Add(x[d, s] == 0)
### OBJECTIVE FUNCTION ###
# want to minimize total distance traveled -> min frequency * distance * binary variable
objective_terms = []
for d in D:
    d_idx = die_index[d]
    for s in S:
        s_idx = shelf_index[s]
        for m in M:
            m_idx = machine_index[m]
            f = freq_matrix[m_idx, d_idx]
            dist = dist_matrix[m_idx, s_idx]
            if f is None or dist is None:
                print(f"Skipping term due to None at m={m}, d={d}, s={s} -> f={f},
dist={dist}")
            else:
                objective_terms.append(x[d, s] * f * dist)

```

```

# minimize
model.Minimize(sum(objective_terms))
# Solve
solver = cp_model.CpSolver()
solver.parameters.max_time_in_seconds = 30 # max time that model can run in seconds
status = solver.Solve(model)
# Output
if status == cp_model.OPTIMAL or status == cp_model.FEASIBLE:
    for d in D:
        for s in S:
            if solver.BooleanValue(x[d, s]):
                print(f'Die {d} assigned to {s}')
    print(f'Total distance: {solver.ObjectiveValue()}')
else:
    print('No solution found.')

```

C Team Contributions

- **Samantha Lynn:** Solution Process; Data Appendices.
- **Stella Beaty:** Economic Analysis and Systems Impacts; Project Schedule; Summary, Conclusions, and Recommendations; Limitations and Reservations.
- **David Colindres:** Executive Summary; Introduction to the Company; Problem Definition; Problem Analysis.